

#5 Quiz-Chapter 14, 15 & 16

Name: _____ Student ID: _____ Instructor: _____ Score: _____

Part A: Multiple choice (Select the one that is best in each case.)

1. The average rate of disappearance of ozone in the reaction $2\text{O}_3(g) \rightarrow 3\text{O}_2(g)$ is found to be 8.8×10^{-3} atm over a certain interval of time. What is the rate of appearance of O_2 during this interval?

A) 5.9×10^{-3} atm/time

B) 2.6×10^{-2} atm/time

C) 1.8×10^{-2} atm/time

D) 1.3×10^{-2} atm/time

E) 8.8×10^{-3} atm/time

2. Consider the following rate law: $\text{Rate} = k[\text{A}]^n[\text{B}]^m$

How are the exponents n and m determined?

A) By using the balanced chemical equation

B) By using the subscripts for the chemical formulas

C) By using the coefficients of the chemical formulas

D) By educated guess

E) By experiment

3. The following data were obtained for the hypothetical reaction $2\text{A} + \text{B} \rightarrow \text{products}$.

$[\text{A}]_0 (M)$	$[\text{B}]_0 (M)$	Initial Rate (M/s)
0.2	0.1	5
0.2	0.2	20
0.6	0.1	15

What is the overall order of this reaction?

A) 4

B) 1/2

C) 0

D) 3

E) 1

4. For a second-order reaction, what are the possible units of the rate constant?

A) $\text{L}^{-1} \cdot \text{s}^{-1}$

B) $\text{mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}$

C) s

D) $\text{L} \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$

E) $\text{mol} \cdot \text{L}^{-1}$

5. A first-order reaction is 58% complete at the end of 11 min. What is the value of the rate constant?

A) 0.079 min^{-1}

B) 0.37 min^{-1}

C) $7.9 \times 10^{-2} \text{ min}^{-1}$

D) $3.4 \times 10^{-2} \text{ min}^{-1}$

E) $5.0 \times 10^{-2} \text{ min}^{-1}$

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6. Of the following, _____ will lower the activation energy for a reaction.

- A) increasing the concentrations of reactants
- B) raising the temperature of the reaction
- C) adding a catalyst for the reaction
- D) removing products as the reaction proceeds
- E) increasing the pressure

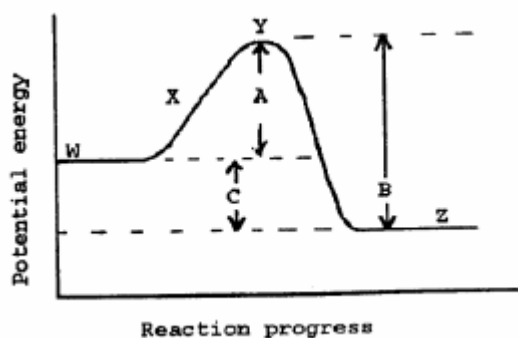
7. The rate law for a reaction is found to be $\text{Rate} = k[\text{A}]^2[\text{B}]$. Which of the following mechanisms gives this rate law?

- I. $\text{A} + \text{B} \rightleftharpoons \text{E}$ (fast)
 $\text{E} + \text{B} \rightarrow \text{C} + \text{D}$ (slow)
 - II. $\text{A} + \text{B} \rightleftharpoons \text{E}$ (fast)
 $\text{E} + \text{A} \rightarrow \text{C} + \text{D}$ (slow)
 - III. $\text{A} + \text{A} \rightarrow \text{E}$ (slow)
 $\text{E} + \text{B} \rightarrow \text{C} + \text{D}$ (fast)
- A) I only **B) II only** C) III only D) two of these E) none of these

8. In general, as temperature goes up, reaction rate _____.

- A) goes up if the reaction is exothermic
- B) goes up if the reaction is endothermic
- C) goes up regardless of whether the reaction is exothermic or endothermic
- D) stays the same regardless of whether the reaction is exothermic or endothermic
- E) stays the same if the reaction is first order

9. The questions below refer to the following diagram:



If the reaction were reversible, would the forward or the reverse reaction have a higher activation energy?

- A) The diagram shows no indication of any activation energy.
- B) The forward and reverse activation energies are equal.
- C) The forward activation energy would be greater.
- D) The reverse activation energy would be greater.
- E) None of these.

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10. Identify the rate equation of the termolecular elementary reaction given below.



- A) $k[A][B][C]$
- B) $k([A]+[B]+[C])$
- C) $k[D][E]$
- D) $k([D]+[E])$
- E) $(k[A][B][C])/(k[D][E])$

11. For which of the following reactions is the ratio K_p/K_c largest at 300 K?

- A) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$
- B) $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$
- C) $Ni(CO)_4(g) \rightleftharpoons Ni(s) + 4CO(g)$
- D) $C(s) + 2H_2(g) \rightleftharpoons CH_4(g)$

12. What is the relationship between the rate constants for the forward and reverse reactions and the equilibrium constant for a process involving just elementary reactions (very simple primary reactions)?

- A) $K = k_f/k_r$
- B) $K = k_f/k_r$
- C) $K = k_f - k_r$
- D) $K = k_f k_r$
- E) $K = k_f + k_r$

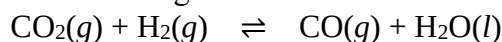
13. Of the following equilibrium, which one will shift to the left in response to a decrease in volume?

- A) $H_2(g) + Cl_2(g) \rightleftharpoons 2HCl(g)$
- B) $4Fe(s) + 3O_2(g) \rightleftharpoons 2Fe_2O_3(s)$
- C) $CO_2(g) + H_2O(l) \rightleftharpoons H_2CO_3(aq)$
- D) $2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$
- E) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

14. How is the reaction quotient used to determine whether a system is at equilibrium?

- A) The reaction quotient must be satisfied for equilibrium to be achieved.
- B) At equilibrium, the reaction quotient is undefined.
- C) The reaction is at equilibrium when $Q < K_{eq}$.
- D) The reaction is at equilibrium when $Q > K_{eq}$.
- E) The reaction is at equilibrium when $Q = K_{eq}$.

15. Which of the following expressions is the correct equilibrium-constant expression for the following reaction?

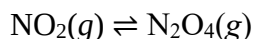


- A) $([CO_2][H_2])/([CO][H_2O])$
- B) $([CO_2][H_2])/[CO]$
- C) $[CO]/([CO_2][H_2])$
- D) $([CO][H_2O])/([CO_2][H_2])$
- E) $[CO_2]/[CO]$

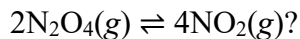
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16. The reaction



has an equilibrium constant of 4.50×10^{-3} at a certain temperature. What is the equilibrium constant of



- A) 4.50×10^3 B) 9.00×10^6 C) 2.20×10^7 D) 2.00×10^7 E) 4.90×10^5

17. Which of the following statements is/are correct?

(i) Every Brønsted-Lowry acid is also a Lewis acid.

(ii) Every Lewis acid is also a Brønsted-Lowry acid.

(iii) Conjugate acids of weak bases produce more acidic solutions than conjugate acids of strong bases.

(iv) K^+ ion is acidic in water because it causes hydrating water molecules to become more acidic.

(v) The percent ionization of a weak acid in water increases as the concentration of acid decreases.

A) (i), (iii) and (v)

B) (i), (ii) and (v)

C) (i) and (iii)

D) (iii) and (v)

E) All of the above.

18. The hydride ion, H^- , is a stronger base than the hydroxide ion, OH^- . The product(s) of the reaction of hydride ion with water is/are _____.

A) H_3O^+ (aq)

B) OH^- (aq) + H_2 (g)

C) OH^- (aq) + 2H^+ (aq)

D) no reaction occurs

E) H_2O_2 (aq)

19. Using the data in the table, which of the conjugate acids below is the strongest acid?

Base	K_b
HS^-	1.1×10^{-7}
CO_3^{2-}	1.8×10^{-4}
ClO^-	3.3×10^{-7}
$(\text{CH}_3)_2\text{NH}$	5.4×10^{-4}

A) H_2S

B) HCO_3^-

C) HClO

D) $(\text{CH}_3)_2\text{NH}_2^+$

E) H_2S and HClO

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20. Which of the following is a Lewis acid?

- A) NH_3 B) OH^- C) Cl^- **D) BF_3** E) CN^-

21. Classify the following compounds as weak acids (W) or strong acids (S):

nitric acid perchloric acid chloric acid

- A) W W W
B) W S S
C) S S S
D) S W W
E) W S W

22. Which of the following aqueous solutions (at 25 °C) has the highest $[\text{OH}^-]$?

- A) a solution with a pH of 6.0
B) a $1 \times 10^{-4} \text{ M}$ solution of HNO_3
C) a solution with a pOH of 12.0
D) pure water
E) a $1 \times 10^{-3} \text{ M}$ solution of NH_4Cl

23. Of the following substances, an aqueous solution of _____ will form basic solutions. NaHS $\text{Cu}(\text{NO}_3)_2$ KHCO_3 NaF

- A) NaHS , $\text{Cu}(\text{NO}_3)_2$
B) KHCO_3 , NaHS
C) NaF only
D) NaF , KHCO_3
E) NaHS , KHCO_3 and NaF

24. Which of the following statements is true?

- A) In general, the acidity of binary acids increases from left to right in a given row of the periodic table.**
B) In a series of acids that have the same central atom, acid strength increases with the number of hydrogen atoms bonded to the central atom.
C) Hydrotelluric acid (H_2Te) is a stronger acid than H_2S because Te is more electronegative than S.
D) For acids of the same general structure but differing electronegativities of the central atoms, acid strength decreases with increasing electronegativity of the central atom.
E) The strongest acid known is HF because fluorine is the most electronegative element.

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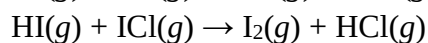
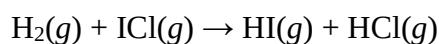
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25. The pH of a 0.55 M aqueous solution of hypobromous acid, HBrO, at 25.0 °C is 4.48. What is the value of K_a for HBrO?

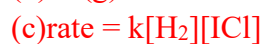
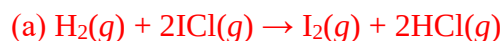
- A) 2.0×10^{-9} B) 1.1×10^{-9} C) 6.0×10^{-5} D) 3.3×10^{-5} E) 3.0×10^4

Part B: Short Answer (Write legibly and show all work for all steps in the problem)

26. The following mechanism has been proposed for the gas-phase reaction of H_2 with ICl:



- (a) Write the balanced equation for the overall reaction.
(b) Identify any intermediates in the mechanism.
(c) If the first step is slow and the second one is fast, which rate law do you expect to be observed for the overall reaction?(3 points)



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27.

- (a) How is the rate at which ozone disappears related to the rate at which oxygen appears in the reaction $2 \text{O}_3(g) \longrightarrow 3 \text{O}_2(g)$?
- (b) If the rate at which O_2 appears, $\Delta[\text{O}_2]/\Delta t$, is $6.0 \times 10^{-5} \text{ M/s}$ at a particular instant, at what rate is O_3 disappearing at this same time, $-\Delta[\text{O}_3]/\Delta t$?

(4 points)

SOLUTION

Analyze We are given a balanced chemical equation and asked to relate the rate of appearance of the product to the rate of disappearance of the reactant.

Plan We can use the coefficients in the chemical equation as shown in Equation 14.4 to express the relative rates of reactions.

Solve

- (a) Using the coefficients in the balanced equation and the relationship given by Equation 14.4, we have:

$$\text{Rate} = -\frac{1}{2} \frac{\Delta[\text{O}_3]}{\Delta t} = \frac{1}{3} \frac{\Delta[\text{O}_2]}{\Delta t}$$

- (b) Solving the equation from part (a) for the rate at which O_3 disappears, $-\Delta[\text{O}_3]/\Delta t$, we have:

$$-\frac{\Delta[\text{O}_3]}{\Delta t} = \frac{2}{3} \frac{\Delta[\text{O}_2]}{\Delta t} = \frac{2}{3} (6.0 \times 10^{-5} \text{ M/s}) = 4.0 \times 10^{-5} \text{ M/s}$$

28.

The initial rate of a reaction $\text{A} + \text{B} \longrightarrow \text{C}$ was measured for several different starting concentrations of A and B, and the results are as follows:

Experiment Number	[A] (M)	[B] (M)	Initial Rate (M/s)
1	0.100	0.100	4.0×10^{-5}
2	0.100	0.200	4.0×10^{-5}
3	0.200	0.100	16.0×10^{-5}

Using these data, determine (a) the rate law for the reaction, (b) the rate constant, (c) the rate of the reaction when $[\text{A}] = 0.050 \text{ M}$ and $[\text{B}] = 0.100 \text{ M}$.

(5 points)

Solve

- (a) If we compare experiments 1 and 2, we see that [A] is held constant and [B] is doubled. Thus, this pair of experiments shows how [B] affects the rate, allowing us to deduce the order of the rate law with

In experiments 1 and 3, [B] is held constant, so these data show how [A] affects rate. Holding [B] constant while doubling [A] increases the rate fourfold. This result indicates that rate is proportional to $[\text{A}]^2$ (that is, the reaction is second order in A). Hence, the rate law is

- (b) Using the rate law and the data from experiment 1, we have

- (c) Using the rate law from part (a) and the rate constant from part (b), we have

respect to B. Because the rate remains the same when [B] is doubled, the concentration of B has no effect on the reaction rate. The rate law is therefore zero order in B (that is, $n = 0$).

$$\text{Rate} = k[\text{A}]^2[\text{B}]^0 = k[\text{A}]^2$$

$$k = \frac{\text{rate}}{[\text{A}]^2} = \frac{4.0 \times 10^{-5} \text{ M/s}}{(0.100 \text{ M})^2} = 4.0 \times 10^{-3} \text{ M}^{-1} \text{ s}^{-1}$$

$$\text{Rate} = k[\text{A}]^2 = (4.0 \times 10^{-3} \text{ M}^{-1} \text{ s}^{-1})(0.050 \text{ M})^2 = 1.0 \times 10^{-5} \text{ M/s}$$

Because [B] is not part of the rate law, it is irrelevant to the rate if there is at least some B present to react with A.

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29. Provide the detailed analysis.

The following data were obtained for the gas-phase decomposition of nitrogen dioxide at 300 °C, $\text{NO}_2(\text{g}) \longrightarrow \text{NO}(\text{g}) + \frac{1}{2}\text{O}_2(\text{g})$. Is the reaction first or second order in NO_2 ?

Time (s)	$[\text{NO}_2]$ (M)
0.0	0.01000
50.0	0.00787
100.0	0.00649
200.0	0.00481
300.0	0.00380

(4 points)

SOLUTION

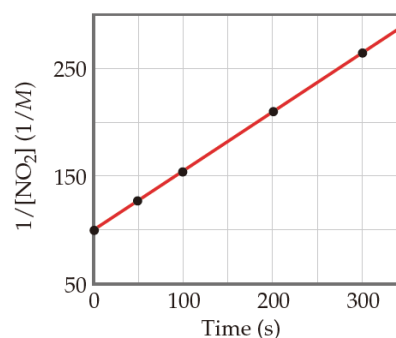
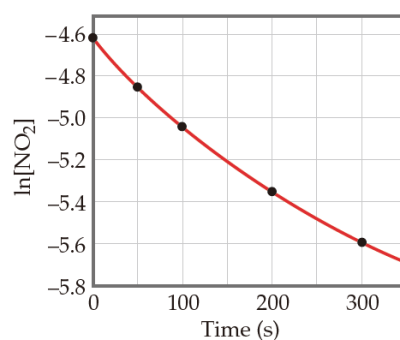
Analyze We are given the concentrations of a reactant at various times during a reaction and asked to determine whether the reaction is first or second order.

Plan We can plot $\ln[\text{NO}_2]$ and $1/[\text{NO}_2]$ against time. If one plot or the other is linear, we will know the reaction is either first or second order.

Solve To graph $\ln[\text{NO}_2]$ and $1/[\text{NO}_2]$ against time, we first make the following calculations from the data given:

Time (s)	$[\text{NO}_2]$ (M)	$\ln[\text{NO}_2]$	$1/[\text{NO}_2]$ (1/M)
0.0	0.01000	-4.605	100
50.0	0.00787	-4.845	127
100.0	0.00649	-5.037	154
200.0	0.00481	-5.337	208
300.0	0.00380	-5.573	263

As ► **Figure 14.9** shows, only the plot of $1/[\text{NO}_2]$ versus time is linear. Thus, the reaction obeys a second-order rate law: $\text{Rate} = k[\text{NO}_2]^2$. From the slope of this straight-line graph, we determine that $k = 0.543 \text{ M}^{-1} \text{ s}^{-1}$ for the disappearance of NO_2 .



▲ **Figure 14.9** Kinetic data for decomposition of NO_2 .

30.

The following table shows the rate constants for the rearrangement of methyl isonitrile at various temperatures (these are the data points in Figure 14.14):

Temperature (°C)	$k (\text{s}^{-1})$
189.7	2.52×10^{-5}
198.9	5.25×10^{-5}
230.3	6.30×10^{-4}
251.2	3.16×10^{-3}

- (a) From these data, calculate the activation energy for the reaction. (b) What is the value of the rate constant at 430.0 K?

(4 points)

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Solve

- (a) We must first convert the temperatures from degrees Celsius to kelvins. We then take the inverse of each temperature, $1/T$, and the natural log of each rate constant, $\ln k$. This gives us the table shown at the right:

T (K)	1/T (K ⁻¹)	ln k
462.9	2.160×10^{-3}	-10.589
472.1	2.118×10^{-3}	-9.855
503.5	1.986×10^{-3}	-7.370
524.4	1.907×10^{-3}	-5.757

A graph of $\ln k$ versus $1/T$ is a straight line (▼ Figure 14.19).

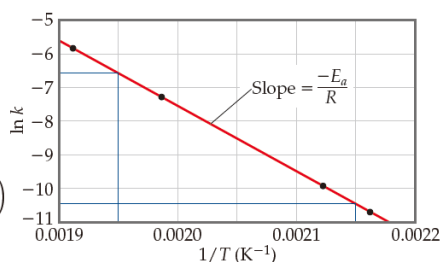
The slope of the line is obtained by choosing any two well-separated points and using the coordinates of each:

$$\text{Slope} = \frac{\Delta y}{\Delta x} = \frac{-6.6 - (-10.4)}{0.00195 - 0.00215} = -1.9 \times 10^4$$

Because logarithms have no units, the numerator in this equation is dimensionless. The denominator has the units of $1/T$, namely, K⁻¹. Thus, the overall units for the slope are K. The slope equals $-E_a/R$. We use the value for the gas constant R in units of J/mol-K (Table 10.2). We thus obtain

$$\begin{aligned} \text{Slope} &= -\frac{E_a}{R} \\ E_a &= -(\text{slope})(R) = -(-1.9 \times 10^4 \text{ K}) \left(8.314 \frac{\text{J}}{\text{mol-K}} \right) \left(\frac{1 \text{ kJ}}{1000 \text{ J}} \right) \\ &= 1.6 \times 10^2 \text{ kJ/mol} = 160 \text{ kJ/mol} \end{aligned}$$

We report the activation energy to only two significant figures because we are limited by the precision with which we can read the graph in Figure 14.19.



▲ Figure 14.19 Graphical determination of activation energy E_a .

- (b) To determine the rate constant, k_1 , at $T_1 = 430.0 \text{ K}$, we can use Equation 14.21 with $E_a = 160 \text{ kJ/mol}$ and one of the rate constants and temperatures from the given data, such as $k_2 = 2.52 \times 10^{-5} \text{ s}^{-1}$ and $T_2 = 462.9 \text{ K}$:

$$\begin{aligned} \ln \left(\frac{k_1}{2.52 \times 10^{-5} \text{ s}^{-1}} \right) &= \\ \left(\frac{160 \text{ kJ/mol}}{8.314 \text{ J/mol-K}} \right) \left(\frac{1}{462.9 \text{ K}} - \frac{1}{430.0 \text{ K}} \right) \left(\frac{1000 \text{ J}}{1 \text{ kJ}} \right) &= -3.18 \end{aligned}$$

Thus,

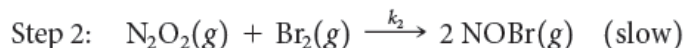
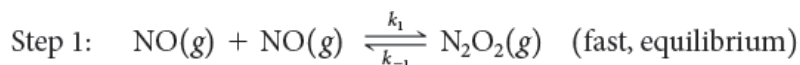
$$\begin{aligned} \frac{k_1}{2.52 \times 10^{-5} \text{ s}^{-1}} &= e^{-3.18} = 4.15 \times 10^{-2} \\ k_1 &= (4.15 \times 10^{-2})(2.52 \times 10^{-5} \text{ s}^{-1}) = 1.0 \times 10^{-6} \text{ s}^{-1} \end{aligned}$$

Note that the units of k_1 are the same as those of k_2 .

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31. Deriving the reaction rate expression for the following reaction. (2 points)



$$\text{Rate} = k_2[\text{N}_2\text{O}_2][\text{Br}_2]$$

$$k_1[\text{NO}]^2 = k_{-1}[\text{N}_2\text{O}_2]$$

$$\text{rate} = k_1 k_2 / k_{-1} [\text{NO}]^2 [\text{Br}_2]$$

32. The decomposition of a certain insecticide in water at 12 °C follows first-order kinetics with a rate constant of 1.45 yr^{-1} . A quantity of this insecticide is washed into a lake on June 1, leading to a concentration of $5.0 \times 10^{-7} \text{ g/cm}^3$. Assume that the temperature of the lake is constant (so that there are no effects of temperature variation on the rate). **(a)** What is the concentration of the insecticide on June 1 of the following year? **(b)** How long will it take for the insecticide concentration to decrease to $3.0 \times 10^{-7} \text{ g/cm}^3$? (4 points)

Solve

(a) Substituting the known quantities into Equation 14.13, we have

$$\ln[\text{insecticide}]_{t=1 \text{ yr}} = -(1.45 \text{ yr}^{-1})(1.00 \text{ yr}) + \ln(5.0 \times 10^{-7})$$

We use the $\ln$ function on a calculator to evaluate the second term on the right [that is, $\ln(5.0 \times 10^{-7})$], giving

$$\ln[\text{insecticide}]_{t=1 \text{ yr}} = -1.45 + (-14.51) = -15.96$$

To obtain $[\text{insecticide}]_{t=1 \text{ yr}}$, we use the inverse natural logarithm, or e^x , function on the calculator:

$$[\text{insecticide}]_{t=1 \text{ yr}} = e^{-15.96} = 1.2 \times 10^{-7} \text{ g/cm}^3$$

Note that the concentration units for $[A]_t$ and $[A]_0$ must be the same.

(b) Again substituting into Equation 14.13, with $[\text{insecticide}]_t = 3.0 \times 10^{-7} \text{ g/cm}^3$, gives

$$\ln(3.0 \times 10^{-7}) = -(1.45 \text{ yr}^{-1})(t) + \ln(5.0 \times 10^{-7})$$

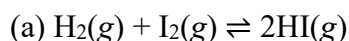
$$t = -[\ln(3.0 \times 10^{-7}) - \ln(5.0 \times 10^{-7})]/1.45 \text{ yr}^{-1}$$

$$= -(-15.02 + 14.51)/1.45 \text{ yr}^{-1} = 0.35 \text{ yr}$$

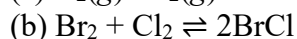
Check In part **(a)** the concentration remaining after 1.00 yr (that is, $1.2 \times 10^{-7} \text{ g/cm}^3$) is less than the original concentration ($5.0 \times 10^{-7} \text{ g/cm}^3$), as it should be. In **(b)** the given concentration

($3.0 \times 10^{-7} \text{ g/cm}^3$) is greater than that remaining after 1.00 yr, indicating that the time must be less than a year. Thus, $t = 0.35 \text{ yr}$ is a reasonable answer.

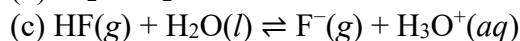
33. List the reactions in order from the greatest extent of reaction to the lowest.



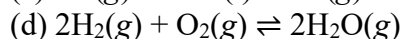
$$K_c = 49$$



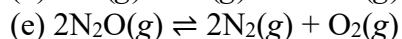
$$K_c = 6.9$$



$$K_c = 6.8 \times 10^{-4}$$



$$K_c = 9.0 \times 10^{80}$$



$$K_c = 7.0 \times 10^{34}$$

(2 points)

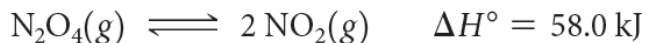
(d)>(e)>(a)>(b)>(c)

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34.

Consider the equilibrium



In which direction will the equilibrium shift when (a) N_2O_4 is added, (b) NO_2 is removed, (c) the pressure is increased by addition of $\text{N}_2(g)$, (d) the volume is increased, (e) the temperature is decreased?

(5 points)

SOLUTION

Analyze We are given a series of changes to be made to a system at equilibrium and are asked to predict what effect each change will have on the position of the equilibrium.

Plan Le Châtelier's principle can be used to determine the effects of each of these changes.

Solve

- (a) The system will adjust to decrease the concentration of the added N_2O_4 , so the equilibrium shifts to the right, in the direction of product.
- (b) The system will adjust to the removal of NO_2 by shifting to the side that produces more NO_2 ; thus, the equilibrium shifts to the right.
- (c) Adding N_2 will increase the total pressure of the system, but N_2 is not involved in the reaction. The partial pressures of NO_2 and N_2O_4 are therefore unchanged, and there is no shift in the position of the equilibrium.
- (d) If the volume is increased, the system will shift in the direction that occupies a larger volume (more gas molecules); thus, the equilibrium shifts to the right.
- (e) The reaction is endothermic, so we can imagine heat as a reagent on the reactant side of the equation. Decreasing the temperature will shift the equilibrium in the direction that produces heat, so the equilibrium shifts to the left, toward the formation of more N_2O_4 . Note that only this last change also affects the value of the equilibrium constant, K .

35. Butyric acid is responsible for the foul smell of rancid butter. The $\text{p}K_a$ of butyric acid is 4.84 at 25 °C. **(4 points)**

(a) Calculate the pH of a 0.050 M solution of butyric acid.

(b) Calculate the pH of a 0.050 M solution of sodium butyrate.

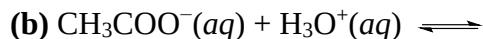
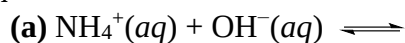
(a)pH = 3.07

(b)pH = 8.77

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36. Predict the products of the following acid–base reactions and whether the equilibrium lies to the left or to the right of the equation. Explain.



(2 points)

(a) right

(b) right

37. Identify conjugate acids and bases

(a) What is the conjugate base of HClO_4 , H_2S , PH_4^+ , HCO_3^- ?

(b) What is the conjugate acid of CN^- , SO_4^{2-} , H_2O , HCO_3^- ?

(4 points)

(a) ClO_4^- , HS^- , PH_3 , CO_3^{2-}

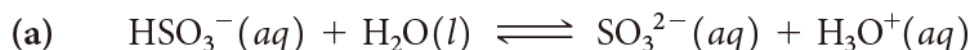
(b) HCN , HSO_4^- , H_3O^+ , H_2CO_3

38. Writing equations for proton-transfer reactions.

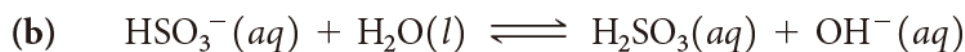
The hydrogen sulfite ion (HSO_3^-) is amphoteric. Write an equation for the reaction of HSO_3^- with water (a) in which the ion acts as an acid and (b) in which the ion acts as a base. In both cases identify the conjugate acid–base pairs.

(2 points)

Solve



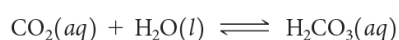
The conjugate pairs in this equation are HSO_3^- (acid) and SO_3^{2-} (conjugate base), and H_2O (base) and H_3O^+ (conjugate acid).



The conjugate pairs in this equation are H_2O (acid) and OH^- (conjugate base), and HSO_3^- (base) and H_2SO_3 (conjugate acid).

39. Calculating the pH of a solution of a polyprotic acid(5 points)

The solubility of CO_2 in water at 25 °C and 0.1 atm is 0.0037 M. The common practice is to assume that all the dissolved CO_2 is in the form of carbonic acid (H_2CO_3), which is produced in the reaction



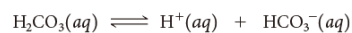
What is the pH of a 0.0037 M solution of H_2CO_3 ?

Also calculate the concentration of $[\text{CO}_3^{2-}]$
 $K_{a1} = 4.3 \times 10^{-7}$, $K_{a2} = 5.6 \times 10^{-11}$

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Solve Proceeding as in Sample Exercises 16.12 and 16.13, we can write the equilibrium reaction and equilibrium concentrations as



Initial concentration (<i>M</i>)	0.0037	0	0
Change in concentration (<i>M</i>)	$-x$	$+x$	$+x$
Equilibrium concentration (<i>M</i>)	$(0.0037 - x)$	x	x

The equilibrium-constant expression is

$$K_{a1} = \frac{[\text{H}^+][\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]} = \frac{(x)(x)}{0.0037 - x} = 4.3 \times 10^{-7}$$

Solving this quadratic equation, we get

$$x = 4.0 \times 10^{-5} M$$

Alternatively, because K_{a1} is small, we can make the simplifying approximation that x is small, so that

$$0.0037 - x \approx 0.0037$$

Thus,

$$\frac{(x)(x)}{0.0037} = 4.3 \times 10^{-7}$$

Solving for x , we have

$$x^2 = (0.0037)(4.3 \times 10^{-7}) = 1.6 \times 10^{-9}$$

$$x = [\text{H}^+] = [\text{HCO}_3^-] = \sqrt{1.6 \times 10^{-9}} = 4.0 \times 10^{-5} M$$

Because we get the same value (to 2 significant figures) our simplifying assumption was justified. The pH is therefore

$$\text{pH} = -\log[\text{H}^+] = -\log(4.0 \times 10^{-5}) = 4.40$$